K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

K.R. MANGALAM UNIVERSITY

For

Bachelor Of Technology

In

Computer Science Engineering Specialization

(Cyber Security)

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Experiment No. 1

Exploring Linux System Commands, Run Levels (Targets), and Help Utilities.

Problem Statement

Efficient use of the Linux operating system requires a strong understanding of basic system commands, system states, and built-in help mechanisms. Unlike graphical operating systems, Linux relies heavily on command-line utilities for system interaction, administration, and automation.

The problem is to explore and practice essential Linux system commands, understand system run levels / targets, and effectively use Linux help utilities such as man, *--help*, and *auto*- completion to access command documentation and usage information.

Aim

To study and execute basic Linux system commands, understand Linux run levels (systemd targets), and use Linux help utilities such as `man`, `--help`, and tab auto-completion.

Theory

Linux Command Line Interface (CLI)

The Linux Command Line Interface allows users to interact with the operating system by typing commands. It provides better control, flexibility, and efficiency compared to graphical interfaces, especially for system administration and automation.

System Run Levels / Targets

- Modern Linux systems use **systemd**, where traditional run levels are replaced by **targets**.

Common systemd targets:

- `graphical.target` – Multi-user mode with GUI
- `multi-user.target` – Multi-user mode without GUI
- `rescue.target` – Single-user rescue mode
- `poweroff.target` – Shutdown system
- `reboot.target` – Reboot system

Linux Help Utilities

Linux provides built-in help tools for learning commands:

- `man` – Displays complete manual pages
- `--help` – Shows brief command usage
- **Tab auto-completion** – Completes commands and file names automatically

Commands Used

Commands	Description
`pwd`	Display present working directory
`ls`	List files and directories
`cd`	Change directory
`whoami`	Display current user
`date`	Show system date and time
`uptime`	Show system running time
`man`	Display manual pages
`--help`	Display help for a command
`systemctl`	Manage system services and targets
`runlevel`	Display previous and current run level
`clear`	Clear terminal screen

Procedure / Steps

Step 1: Login:

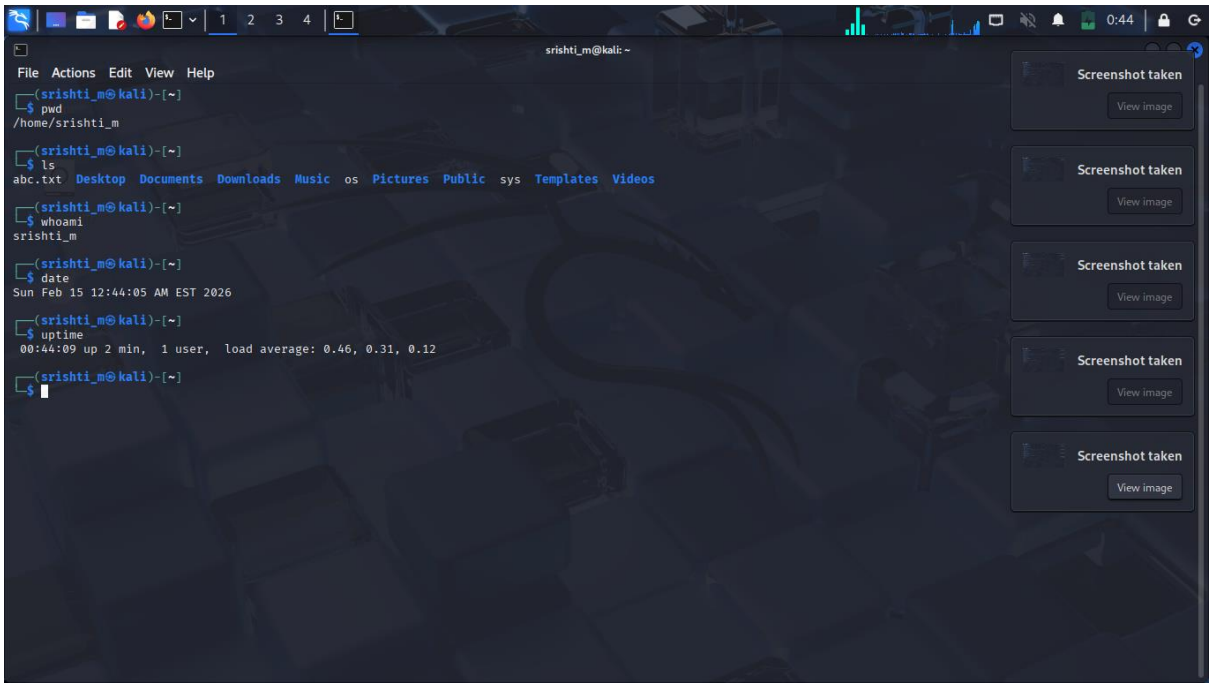
Log in to the Linux virtual machine and open the Terminal.

Step 2: Execute Basic Linux Commands in the Terminal.

Commands:

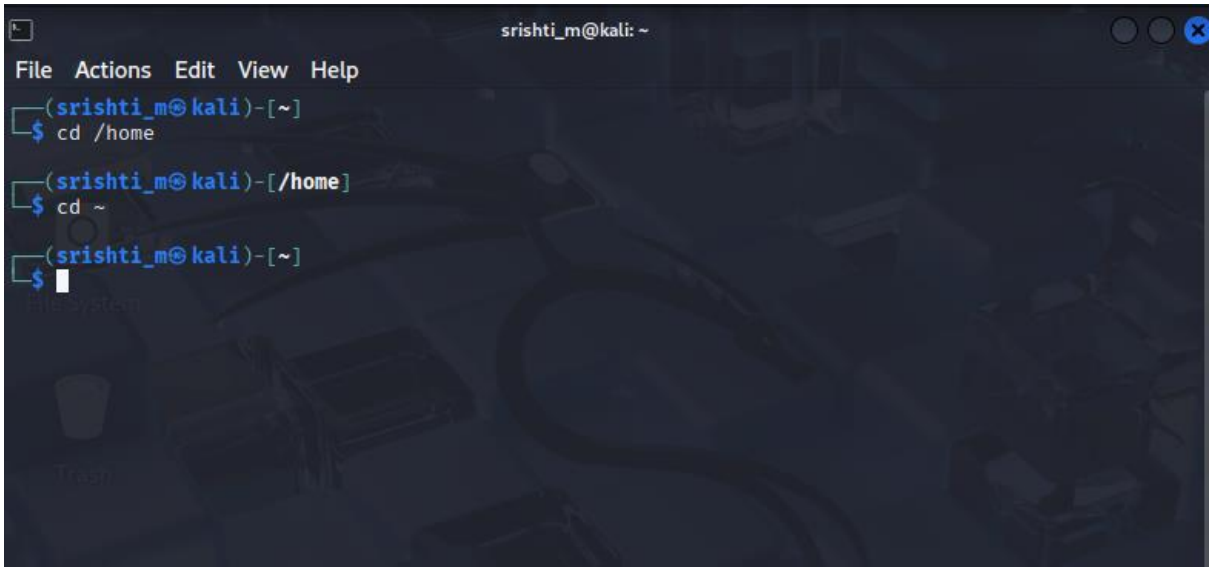
- `pwd` shows current directory
- `ls` lists files
- `whoami` displays logged-in user
- `date` shows system date/time

- `uptime` shows how long system is running



Step 3: Directory Navigation using:

- `cd /home` moves to home directory.
- `cd ~` returns to user's home directory.

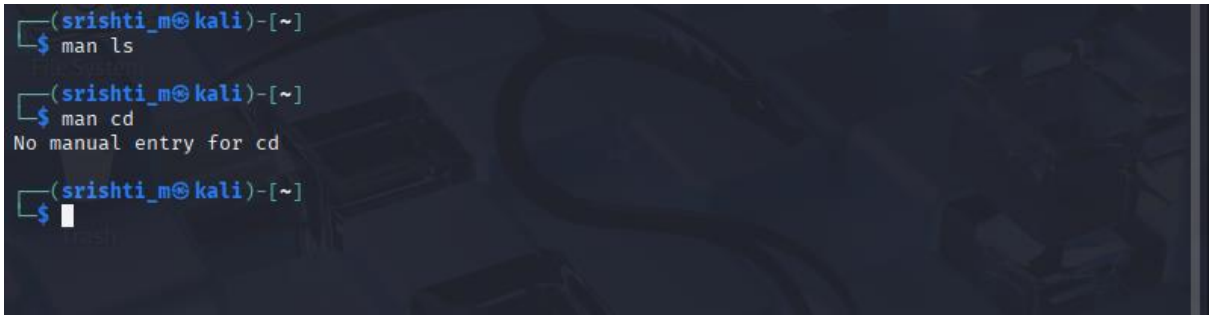


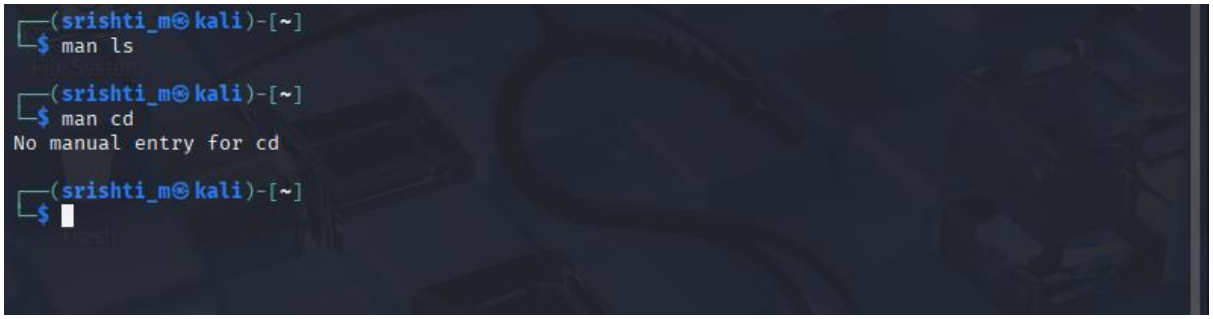
Step 4: Using Manual Pages

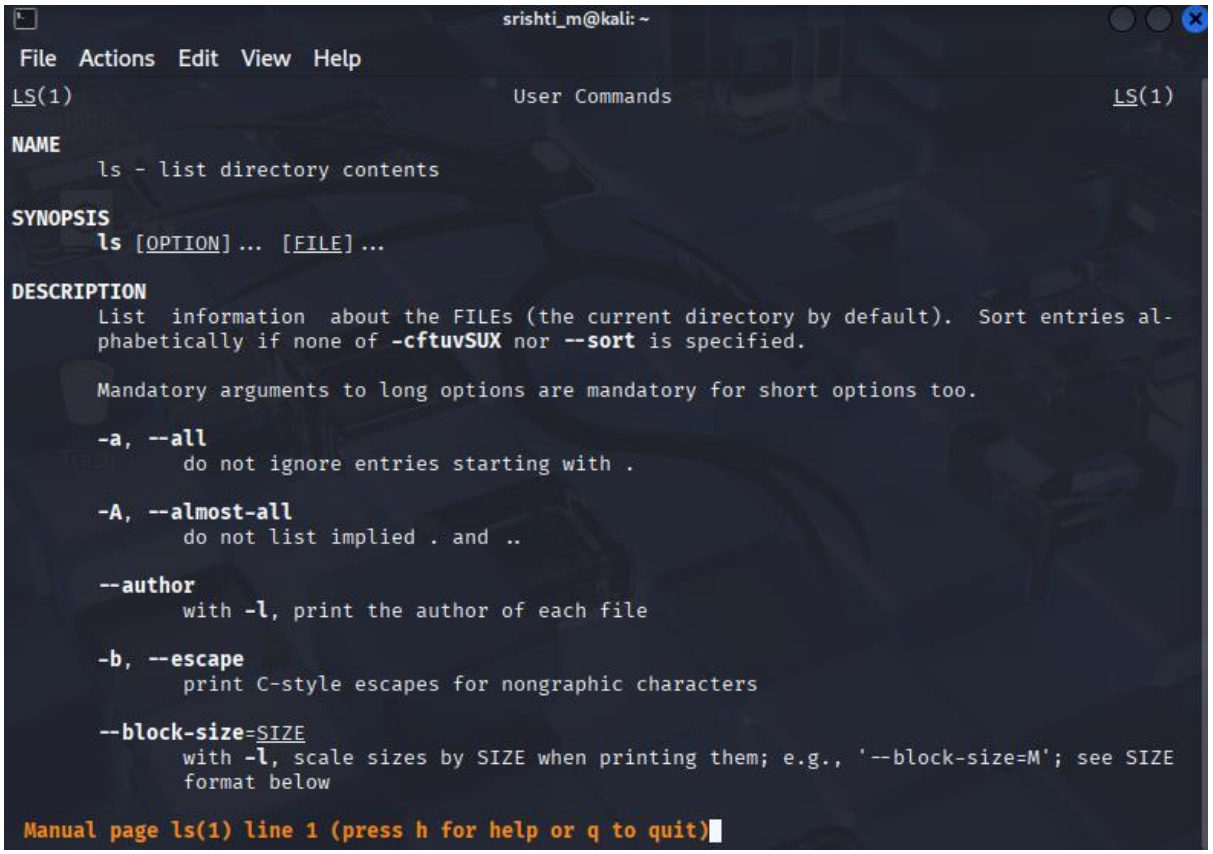
- `man ls`
- `man cd`

It Read command details.

➢ Press `q` to exit the manual.







Step 5: Using Help Option:

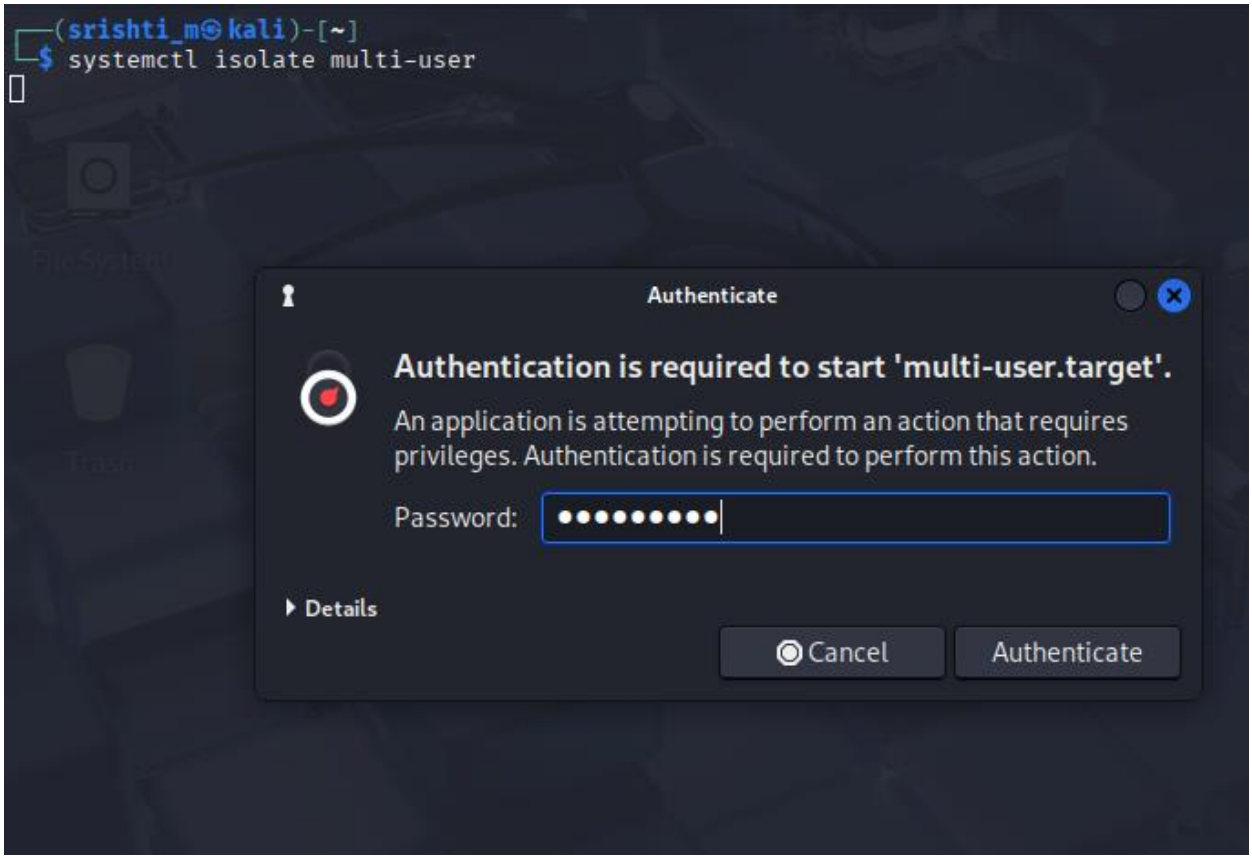
- `ls -help`
- `systemctl -help`

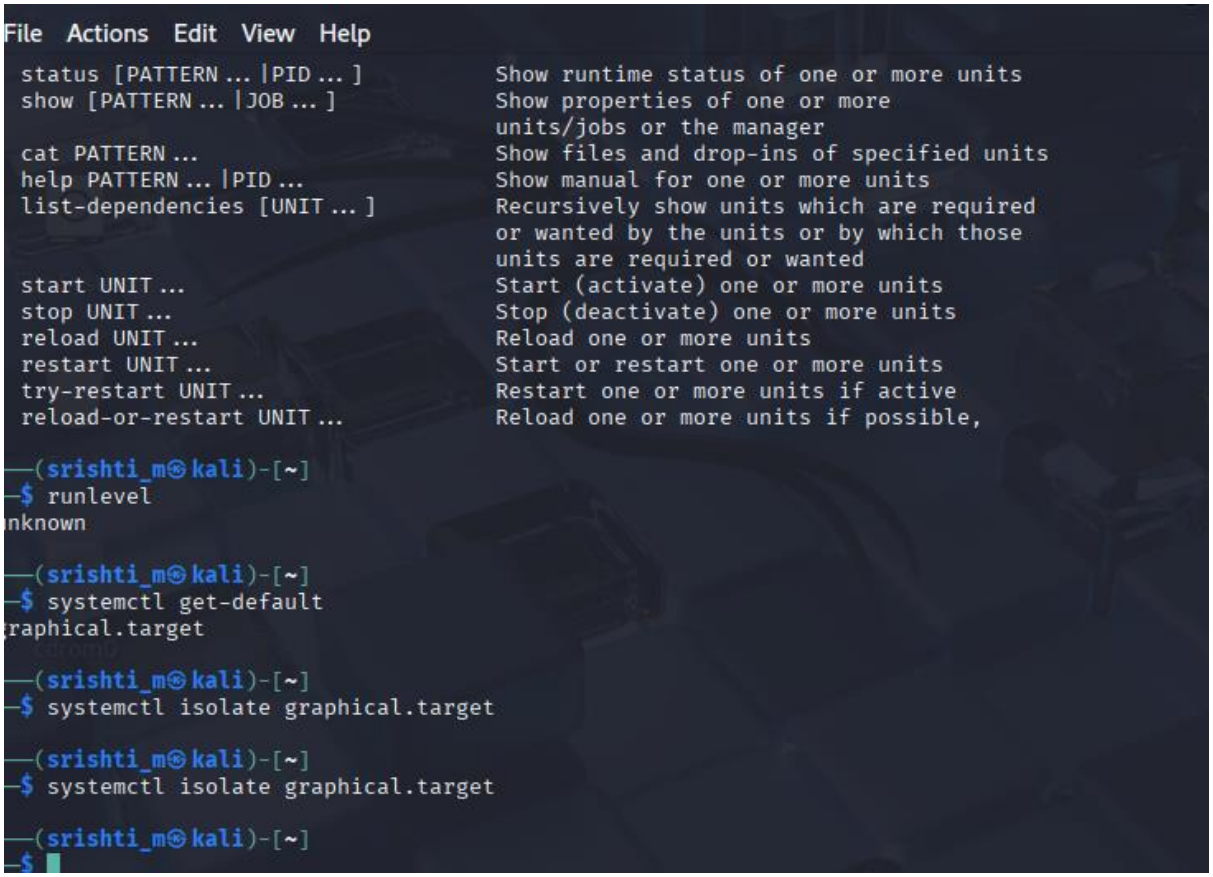
  This displays short usage instructions and optio

```
File  Actions  Edit  View  Help
┌──(srishti_m㉿kali)-[~]
└─$ man cd
No manual entry for cd

┌──(srishti_m㉿kali)-[~]
└─$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                             e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                             change of file status information);
                             with -l: show ctime and sort by name;
                             otherwise: sort by ctime, newest first

  -C                         list entries by columns
      --color[=WHEN]         color the output WHEN; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         same as -a -U
  -F, --classify[=WHEN]      append indicator (one of */=>@|) to entries WHEN
      --file-type            likewise, except do not append '*'
File  Actions  Edit  View  Help

┌──(srishti_m㉿kali)-[~]
└─$ systemctl --help
systemctl [OPTIONS...] COMMAND ...

Query or send control commands to the system manager.

Unit Commands:
  list-units [PATTERN...]             List units currently in memory
  list-automounts [PATTERN...]        List automount units currently in memory,
                                      ordered by path
  list-paths [PATTERN...]             List path units currently in memory,
                                      ordered by path
  list-sockets [PATTERN...]           List socket units currently in memory,
                                      ordered by address
  list-timers [PATTERN...]            List timer units currently in memory,
                                      ordered by next elapse
  is-active PATTERN...                Check whether units are active
  is-failed [PATTERN...]              Check whether units are failed or
                                      system is in degraded state
  status [PATTERN...|PID...]          Show runtime status of one or more units
  show [PATTERN...|JOB...]            Show properties of one or more
                                      units/jobs or the manager
  cat PATTERN...                      Show files and drop-ins of specified units
  help PATTERN...|PID...              Show manual for one or more units
  list-dependencies [UNIT...]         Recursively show units which are required
                                      or wanted by the units or by which those
                                      units are required or wanted
  start UNIT...                       Start (activate) one or more units
  stop UNIT...                        Stop (deactivate) one or more units
  reload UNIT...                      Reload one or more units
```

Step 6: Check System Run Level / Target

- `runlevel`- "Kali Linux uses systemd, so traditional runlevel commands are deprecated and `systemctl` targets are used instead."
- `systemctl get-default` shows default system target.


(srishti_m㉿kali)-[~]
$ systemctl isolate multi-user
Authenticate
Authentication is required to start 'multi-user.target'.
An application is attempting to perform an action that requires privileges. Authentication is required to perform this action.
Password:
Details
Cancel
Authenticate


File Actions Edit View Help
status [PATTERN ... |PID ...] Show runtime status of one or more units
show [PATTERN ... |JOB ...] Show properties of one or more units/jobs or the manager
cat PATTERN ... Show files and drop-ins of specified units
help PATTERN ... |PID ... Show manual for one or more units
list-dependencies [UNIT ...] Recursively show units which are required or wanted by the units or by which those units are required or wanted
start UNIT ... Start (activate) one or more units
stop UNIT ... Stop (deactivate) one or more units
reload UNIT ... Reload one or more units
restart UNIT ... Start or restart one or more units
try-restart UNIT ... Restart one or more units if active
reload-or-restart UNIT ... Reload one or more units if possible,
(srishti_m㉿kali)-[~]
$ runlevel
nknown
(srishti_m㉿kali)-[~]
$ systemctl get-default
raphical.target
(srishti_m㉿kali)-[~]
$ systemctl isolate graphical.target
(srishti_m㉿kali)-[~]
$ systemctl isolate graphical.target
(srishti_m㉿kali)-[~]
$

Step 7:- Practice auto-completion by typing partial command names and pressing Tab.

Step 8 :- Clear the terminal screen: Clear

Outcomes

- Successful execution of basic Linux commands
- Display of manual pages and help options
- Identification of current run level/target
- Improved command-line efficiency

Result

Thus, basic Linux system commands, system run levels, and help utilities were successfully explored. The student gained practical understanding of Linux command-line interaction and Linux documentation tools.

Experiment No. 2

Title

File and Directory Management and File Permissions in Linux

Aim

To study and practice Linux commands for file and directory management, understand Linux file and directory permissions, and create and edit files using text editors such as `nano` and `vi`.

Theory

Linux File System

Linux follows a hierarchical file system starting from the root directory `/`. All files and directories are arranged in a tree structure, allowing efficient organization and secure access control.

Files and Directories

- **Files** store data such as text, programs, or scripts.
- **Directories** store files and subdirectories.

File and Directory Permissions

Linux uses permissions to control access:

- **Read (r)** – Read file contents / list directory
- **Write (w)** – Modify file or directory
- **Execute (x)** – Execute file or access directory

Permissions apply to:

- **Owner**
- **Group**
- **Others**

Text Editors

- **nano:** Simple and beginner-friendly text editor
- **vi:** Powerful editor with command and insert modes

Commands Used

Commands	Description
pwd	Display current directory
ls	List files and directories
ls -l	Display detailed file information
cd	Change directory
mkdir	Create directory
rmdir	Remove empty directory
touch	Create empty file
cat	Display file contents
nano	Edit files using nano editor
vi	Edit files using vi editor
chmod	Change file permissions
stat	Display file status

Procedure / Steps

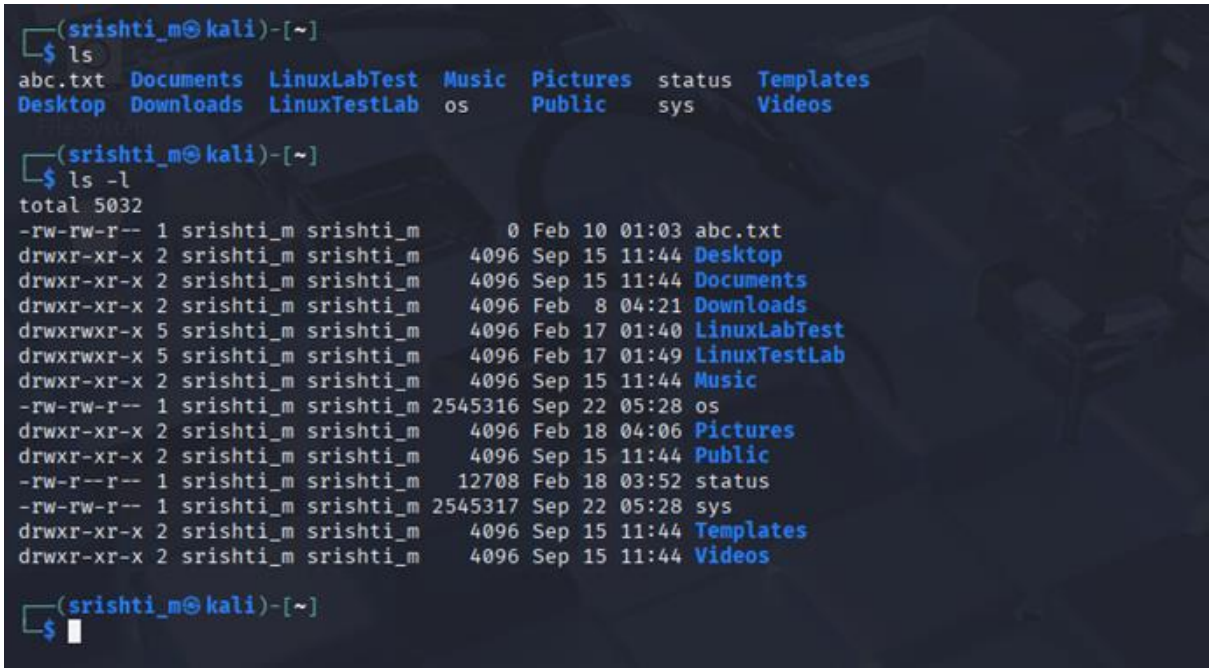
Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Display the Current Directory using `pwd command`.

Step 3: List Files and Directories using:

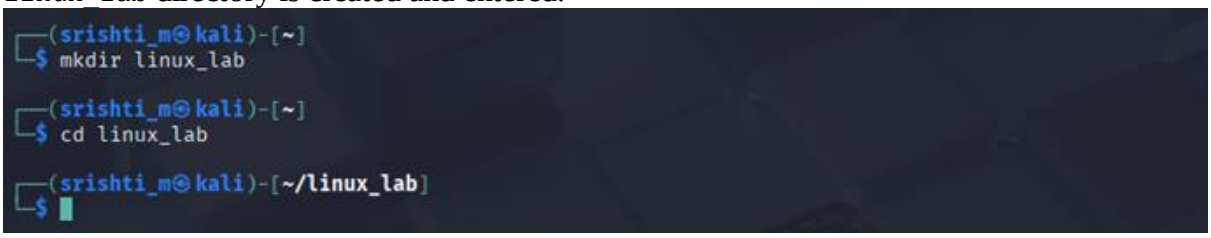
- `ls`
- `ls -l`



Step 4: Create a New Directory

```
mkdir linux_lab
cd linux_lab
```

- `linux_lab` directory is created and entered.



Step 5: Create Files using : `-touch file1.txt file2.txt`

Step 6: Edit File Using : `nano file1.txt`

- Type some text
- Press **Ctrl + O** → Enter (save)
- Press **Ctrl + X** (exit)

Step 7: Edit File Using : `vi file2.txt`

```
┌──(srishti_m㉿kali)-[~]
└─$ mkdir linux_lab2

┌──(srishti_m㉿kali)-[~]
└─$ cd linux_lab2

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ touch file1.txt file2.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ nano file1.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ vi file2.txt
```

Then,

- Press **i** to enter insert mode
- Type text
- Press **Esc**
- Type: `:wq`
- Press Enter

```
srishti_m@kali: ~/linux_lab
File  Actions  Edit  View  Help
linux is powerful
linux is open source
~
~
~
```

Step 9: Check File Permissions

- `ls -l`

```
┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ cat file1.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ ls -l
total 8
-rw-rw-r-- 1 srishti_m srishti_m  0 Feb 20 02:59 file1.txt
-rw-rw-r-- 1 srishti_m srishti_m 55 Feb 20 03:01 file1.txtn
-rw-rw-r-- 1 srishti_m srishti_m 51 Feb 20 03:07 file2.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$
```

Step 10: Change File Permissions using:

- `chmod 755 file1.txt`

```
┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ ls -l
total 8
-rw-rw-r-- 1 srishti_m srishti_m  0 Feb 20 02:59 file1.txt
-rw-rw-r-- 1 srishti_m srishti_m 55 Feb 20 03:01 file1.txtn
-rw-rw-r-- 1 srishti_m srishti_m 51 Feb 20 03:07 file2.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ chmod 755 file1.txt
```

Step 11: Verify Updated Permissions Using:

- `ls -l file1.txt`
- we will observe permission format: `-rw-r--r–`

```
┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ chmod 755 file1.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$ ls -l file1.txt
-rwxr-xr-x 1 srishti_m srishti_m 0 Feb 20 02:59 file1.txt

┌──(srishti_m㉿kali)-[~/linux_lab2]
└─$
```

Outcomes

- ✓ Successful creation of directories and files
- ✓ Files edited using `nano` and `vi`
- ✓ File contents displayed using `cat`
- ✓ File permissions viewed and modified using `chmod`

Result

- ❖ Thus, file and directory management commands were successfully executed in Linux. The student learned to create, edit, and manage files and directories, understand file permissions, and use text editors such as `nano` and `vi`.